AAD-50

USER MANUAL

Windows GUI Application

PRODUCT	:	The Abeselom ASIC-Direct 50 (AAD-50)
VERSION	:	1.1
PLATFORM	:	Windows 10 1607+ / Windows 11
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REPOSITORY	:	github.com/yonasabeselom/aad50
DATE	:	June 2026
LICENCE	:	Proprietary -- Personal use only

This document provides complete operating instructions for the AAD-50 Windows GUI application. Read Section 11 (Safety) before performing any live data destruction operation.

NEW IN v1.1 -- USB ENCLOSURE SUPPORT
AAD-50 now supports NVMe drives in USB 3.x enclosures (UASP).
Three-tier passthrough auto-detection: NVMe Direct / ATA SAT / Storage Reinitialize.
Active pathway is shown in telemetry stream and recorded in audit report.

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SECTION 01 === INTRODUCTION

AAD-50 is an open-source, firmware-enforced NVMe solid-state drive sanitization framework. It bypasses the operating system entirely and communicates directly with the drive controller ASIC via raw IOCTL commands, reaching physical NAND cells invisible to standard tools.

WHY STANDARD TOOLS FAIL ON SSDs

```
Problem: DoD 5220.22-M / Gutmann 35-pass
----------------------------------------
Designed for magnetic HDDs. Intercepted by the
Flash Translation Layer (FTL) on NVMe SSDs.

67% of SSD data survives standard overwrite.
(University of California San Diego, 2011)

AAD-50 bypasses the FTL entirely.
```

DESTRUCTION MATRIX

PHASE	CYCLES	ACTION	CDW10
B	01-40	Physical NAND Cell Overwrite	0x02
C	41-45	Flash Translation Layer Reset	0x01
A	46-50	Cryptographic Key Destruction	0x04

Each cycle is hardware-confirmed via Log Page 0x81 (SSTAT) polling.

KNOWN LIMITATIONS & ROADMAP

```
v1.1 does not yet set the NDAS bit (CDW11 bit 9); a future
version will enforce immediate physical block deallocation.
The 40-cycle Phase B count is a provisional engineering
margin, not a figure derived from cited research; see README.
Full limitations and roadmap: see README.md in the repository.

Plain-language background: The_Unaudited_Drive.pdf
```

SECTION 02 === SYSTEM REQUIREMENTS

REQUIREMENT	VALUE	NOTES
Operating System	Windows 10 v1607+	Win 11 supported
Privileges	Administrator	Required
Drive Type (Direct)	NVMe SSD	M.2 / PCIe slot
Drive Type (USB)	NVMe in USB enclosure	UASP required (v1.1)
Python (optional)	3.10+	Not needed for .exe
Disk Space	50 MB	App + reports

SECTION 03 === INSTALLATION

OPTION A -- STANDALONE EXE (RECOMMENDED)

No Python required. Runs on any Windows 10/11 machine.

[1] Go to: github.com/yonasabeselom/aad50/releases
[2] Download AAD50.exe from the latest release
[3] Right-click AAD50.exe --> Run as administrator

OPTION B -- PYTHON SCRIPT

```
C:\> pip install customtkinter reportlab
C:\> cd C:\path\to\AAD50
C:\AAD50> python aad50_gui_windows.py
```

NOTE: aad50_gui_windows.py and aad50_abeselom_windows.py
must be in the same folder.

SECTION 04 === GETTING STARTED -- FIRST RUN

Complete walkthrough for your first simulation run.
Dry-Run mode is ON by default. Your drive is safe.

[01] Launch	Right-click AAD50.exe --> Run as administrator
[02] Verify safe	Header must show: SIMULATION ACTIVE (amber)
[03] Select Drive	Click 'Select Drive' in the left sidebar
[04] Choose target	Click 'Select Drive' next to your NVMe drive
[05] Continue	Click 'Continue to Sanitize' button
[06] Operator Name	Type your name in the Operator Name field
[07] Authorize	Type exactly: EXECUTE-AAD-50-ABESELOM
[08] Execute	Click 'RUN SIMULATION (DRY-RUN -- SAFE)'
[09] Results	View completion screen with SHA-256 audit hash
[10] Save report	Click 'PDF Certificate' to save your certificate

SECTION 05 === THE FIVE SCREENS

[HOME DASHBOARD]

Main overview screen.
Shows: phase matrix, 50 cycles, 3 phases, opcode 0x84, SHA-256.
Click 'Start -- Select Your Drive' to begin.

[SELECT DRIVE]

Detects all NVMe drives automatically -- including USB enclosures [USB].
Shows: drive model, device path, volume letters, bus type.
First scan: 1-2 seconds. Subsequent: instant (cached).
Click 'Select Drive' then 'Continue to Sanitize'.
USB drives show [USB] tag -- tier detection runs automatically.

[SANITIZE DRIVE]

Main operational screen.
Left: warning description (simulation=amber / live=red).
Right: drive info, operator name, auth token, execute.
Dry-Run toggle in sidebar controls mode.

[AUDIT REPORTS]

Load and verify saved JSON audit reports.
Recalculates SHA-256 hash from cycle records.
VERIFIED = report is genuine and unaltered.
FAILED = report was modified after generation.

[ABOUT]

Tool info, version, author, compliance standards.
Licence information and contact details.
OEM Driver Diagnostic: tests Log Page 0x81 pass-through.

SECTION 06 === DRY-RUN SIMULATION MODE

Test AAD-50 safely. Zero hardware commands sent.

WHAT SIMULATION MODE DOES:

[+] Runs all 50 cycles -- identical timing and logging to live
[+] Skips all DeviceIoControl IOCTL calls completely
[+] Generates real SHA-256 audit hash from simulated records
[+] Never opens drive handle -- OS, files, drive untouched
[+] Produces full JSON report and PDF Certificate
[+] Pathway shown as NONE (no device probed in simulation)

MODE INDICATORS:

LOCATION	SIMULATION MODE	LIVE MODE
Header badge	SIMULATION ACTIVE	REAL WIPE ARMED
Badge color	amber	red
Sidebar label	Simulation active -- safe mode	LIVE MODE -- DATA WILL BE DESTROYED
Execute button	RUN SIMULATION (DRY-RUN -- SAFE)	EXECUTE AAD-50 SANITIZATION -- LIVE

SECTION 07 === LIVE SANITIZATION -- REAL HARDWARE

/!\ WARNING

LIVE MODE PERMANENTLY AND IRREVERSIBLY DESTROYS ALL DATA.
NO RECOVERY IS POSSIBLE.
ONLY PROCEED IF YOU ARE ABSOLUTELY CERTAIN.

PRE-FLIGHT CHECKLIST:

[ ] Target drive contains NO data you want to keep
[ ] Target drive is NOT your Windows boot drive (C:)
[ ] Running as Administrator
[ ] Operator Name ready to enter
[ ] Save location ready for audit report

EXECUTION STEPS:

[01] Toggle Dry-Run OFF in sidebar (switch turns grey)
[02] Verify: header shows REAL WIPE ARMED [DESTRUCTIVE]
[03] Click Select Drive
[04] Confirm correct drive -- check model AND serial number
[05] Click Continue to Sanitize
[06] Read the full IRREVERSIBLE DATA DESTRUCTION WARNING
[07] Enter operator name
[08] Type: EXECUTE-AAD-50-ABESELOM
[09] Click EXECUTE AAD-50 SANITIZATION -- LIVE
[10] Confirm second popup -- click Yes
[11] Monitor 50-cycle progress -- tier shown in telemetry stream
[12] Save PDF Certificate on completion screen

SECTION 08 === USB ENCLOSURE SUPPORT (v1.1)

[v1.1]

AAD-50 v1.1 supports NVMe drives connected via USB 3.x enclosures with UASP support. The tool automatically probes the device at runtime and selects the highest available command pathway before the first cycle.

THREE-TIER AUTO-DETECTION

TIER 1 -- NVMe Direct

IOCTL: IOCTL_STORAGE_PROTOCOL_COMMAND (0x0002D14C)
Full Log Page 0x81 hardware confirmation active.
Works for: M.2/PCIe drives + UASP enclosures with NVMe passthrough.
Chips: ASMedia ASM2364, Realtek RTL9210B.
Polling: hardware-confirmed per cycle.

TIER 2 -- ATA SCSI Passthrough

IOCTL: IOCTL_ATA_PASS_THROUGH (0x0004D02C)
ATA SANITIZE DEVICE (0xB4) via SCSI/ATA Translation (SAT).
Works for: USB enclosures supporting UASP + SAT.
Phase B/C/A map to ATA feature codes 0x0011, 0x0012, 0x0014.
Polling: time-based wait (120s per cycle).

TIER 3 -- Storage Reinitialize

IOCTL: IOCTL_STORAGE_REINITIALIZE_MEDIA (0x0002D504)
Windows storage stack sanitize trigger.
Works for: enclosures blocking NVMe and ATA passthrough.
Polling: time-based wait (180s per cycle).

PATHWAY IN AUDIT REPORT:

The active tier is stored in 'pathway_used' in the JSON report and shown in the telemetry stream for every cycle.

IF TIER = NONE: USB bridge blocking all pathways.
Solution: Install drive directly in M.2 slot and retry.

SECTION 09 === AUDIT REPORTS

Every run produces a JSON audit report sealed with SHA-256.

JSON REPORT FIELDS:

FIELD	DESCRIPTION
tool	Tool name and version
device	Drive path e.g. \\.\PhysicalDrive0
drive_model	Manufacturer and model
drive_serial	Drive serial number
operator	Operator name entered at runtime
started_at	UTC start timestamp
completed_at	UTC completion timestamp
dry_run	true (simulation) or false (live)
pathway_used	NVMe-Direct / ATA-Passthrough / Storage-Reinitialize / None
cycles	Array of 50 cycle records (cycle, phase, action_code,
	timestamp, status, pathway, duration_sec)
outcome	SUCCESS -- DATA DESTROYED or FAILED
log_hash	SHA-256 hash of all cycle records

VERIFYING A REPORT:

[1] Click 'Audit Reports' in the sidebar
[2] Click 'Load JSON Report'
[3] Select your .json file
[4] Check result: VERIFIED = genuine | FAILED = tampered

SECTION 10 === PDF CERTIFICATE OF DESTRUCTION

Professional compliance document for GDPR, HIPAA, ISO 27001 audits.

REQUIREMENT:

```
C:\> pip install reportlab
```

CERTIFICATE CONTENTS:

```
[+] Outcome banner (green=SUCCESS / red=FAILED)
[+] Drive model, serial number, device path, volume letters
[+] All 3 phases with cycle counts and timestamps
[+] Operator name, execution mode, and pathway used
[+] NIST SP 800-88 Rev.2 / IEEE 2883-2022 / ISO/IEC 27040
[+] SHA-256 audit hash (bright green on dark background)
[+] Author, GitHub link, validity statement
```

HOW TO GENERATE:

[1] Complete a sanitization run (simulation or live)
[2] On completion screen: click 'PDF Certificate'
[3] Save dialog defaults to Documents folder
[4] Open PDF and verify serial number, operator, hash

SECTION 11 === TROUBLESHOOTING

```
PROBLEM : No drives appear
CAUSE   : Not running as Administrator.
FIX     : Right-click --> Run as administrator
```

```
PROBLEM : NameError on launch
CAUSE   : Outdated .py file.
FIX     : Download latest from github.com/yonasabeselom/aad50
```

```
PROBLEM : PDF generation fails
CAUSE   : ReportLab not installed.
FIX     : pip install reportlab
```

```
PROBLEM : PDF permission denied
CAUSE   : Saving to restricted location.
FIX     : Save to Documents folder
```

```
PROBLEM : Drive scan slow (1-2s)
CAUSE   : Normal on first visit -- cached after.
FIX     : Expected. Only refresh when drives change.
```

```
PROBLEM : Auth token rejected
CAUSE   : Wrong capitalisation or hyphens.
FIX     : Type: EXECUTE-AAD-50-ABESELOM exactly
```

```
PROBLEM : Driver test fails
CAUSE   : NVMe driver lacks Log Page 0x81 support.
FIX     : Update NVMe driver from manufacturer website
```

```
PROBLEM : USB drive not in list
CAUSE   : Drive not detected as USB-attached NVMe.
FIX     : Ensure enclosure uses UASP. Check Device Manager.
```

```
PROBLEM : Pathway shows NONE
CAUSE   : USB bridge blocking all sanitize pathways.
FIX     : Remove from enclosure. Install in M.2 slot directly.
```

```
PROBLEM : Cycles take 2-3 min (USB)
CAUSE   : Tier 2/3 uses time-based wait (120-180s per cycle).
FIX     : Expected. Total ~100-150 min. Do not interrupt.
```

SECTION 12 === SAFETY AND WARNINGS

/!\ WARNING

READ THIS SECTION BEFORE PERFORMING ANY LIVE SANITIZATION. DATA LOSS IS PERMANENT.

ABSOLUTE RULES:

[!] NEVER run live mode on a drive with data you want to keep
[!] NEVER run live mode on your Windows boot drive (C:)
[!] ALWAYS verify drive model AND serial number before executing
[!] ALWAYS run dry-run simulation first on any new machine
[!] ALWAYS save audit report and PDF certificate after execution
[!] ALWAYS run as Administrator

COMPLIANCE STANDARDS:

STANDARD	RELEVANCE
NIST SP 800-88 Rev.2	Purge class for solid-state media
IEEE 2883-2022	Storage device sanitization
ISO/IEC 27040:2015	Storage security guidelines
GDPR Article 17	Right to erasure
HIPAA	Medical data disposal requirements

END OF DOCUMENT
github.com/yonasabeselom/aad50

